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SUPERSEDING
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MILITARY SPECIFICATION

AIR CLEANERS, INTAKE: DRY-TYPE

(FOR INTERNAL-COMBUSTION ENGINE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers intake air cleaners, including elements of the two-stage dry type for use with internal-combustion engines in stationary applications, industrial and construction equipment, and other specialized applications.

1.2 Classification. The air cleaners shall be of the following classes, as specified (see 6.2):

Class I - 8 hours, 70 percent primary efficiency.
Class II - 12 hours, 70 percent primary efficiency.
Class III - 15 hours, 90 percent primary efficiency (design optional).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-F-506	- Flux, Soldering; Paste and Liquid.
QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy and Lead Alloy.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2940

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS39247	- Air Cleaner, Intake; Dry Type, for Internal Combustion Engines (Class I and II).

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Class I and class II air cleaners shall be as shown on MS39247 and as specified herein. Class III air cleaners shall be as specified herein.

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3.2 First article (first-produced air cleaners and elements). The contractor shall furnish eight or more air cleaners and elements for examination and test within the time frame specified (see 6.2), to prove that his production methods will produce air cleaners and elements that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractors and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Solder. Solder shall conform to QQ-S-571.

3.3.1.1 Soldering flux. Soldering flux shall conform to O-F-506.

3.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for servicing, repair, and replacement without the use of special tools and without removing or disturbing the clean-air chamber connection to the engine. The number of parts required to be removed for servicing shall be held to a minimum. Parts shall be fabricated so as to prevent incorrect reassembly. Covers or plates which must be removed for adjustment, repair, replacement, or servicing shall be equipped with quick-disconnect fastenings.

3.4.1 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.5 Tubular joints. Tubular joints, at the outlet only, shall incorporate an externally beaded end that shall withstand external pressure caused by hose clamps or other fastening devices.

3.6 Maximum rated airflow. Unless otherwise specified (see 6.2), the maximum rated airflow capacity in cubic feet per minute of air shall be as specified in MS39247.

3.7 Performance characteristics. Air cleaners shall have the performance characteristics specified herein.

3.7.1 Resistance to air leakage. The clean-air chamber of an air cleaner shall not leak under 2 pounds per square inch air pressure or when subjected to a vacuum of 50 plus or minus 1-1/2 inches of water.

3.7.2 Air cleaner restriction. The restriction (see 6.4.1) of a new air cleaner at maximum rated airflow shall be not more than 8 inches of water for class I air cleaners, and 12 inches of water for class II and class III air cleaners.

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3.7.3 Air cleaner efficiency. When tested as specified in 4.5.2.5, the air cleaner efficiency shall be not less than 99 percent.

3.7.4 Dust capacity. When tested as specified in 4.5.2.6, the class I air cleaner shall operate for 8 hours, the class II air cleaner for 12 hours, and the class III air cleaner for 15 hours under constant flow with no component servicing. Operation shall be based on 20 inches of water restriction.

3.7.4.1 Air cleaner efficiency during dust-capacity test. When tested as specified in 4.5.2.6, the accumulative efficiency (see 6.4.2) to 20 inches of water shall be not less than 99.5 percent. The initial and second efficiency shall be not less than 99.5 percent. The accumulative efficiency from 20 to 50 inches of water or 10 hours, whichever occurs first, shall be not less than 99 percent. The specific efficiency (see 6.4.3) at 50 inches of water or 10 hours, whichever occurs first, shall be not less than 99 percent.

3.7.4.2 Air cleaner primary separator efficiency. When tested as specified in 4.5.2.6, the primary separator efficiency shall be not less than 70 percent for classes I and II air cleaners, and not less than 90 percent for class III air cleaners.

3.7.5 Service life. When tested as specified in 4.5.2.7, the service life of class I air cleaner shall be 15 hours with a frequency of element cleaning of not less than 1 hour intervals. The service life of class II and class III air cleaners shall be 25 hours and 30 hours respectively, with a frequency of element cleaning of not less than 1-1/2-hour intervals. The efficiency of any air cleaner shall be not less than 99.5 percent during any cycle. The arithmetic average of the primary separator efficiency shall be not less than 70 percent for classes I and II and 90 percent for class III air cleaners.

3.7.6 Restriction after first cycle of service test. The restriction at maximum rated airflow of the air cleaners after one servicing shall be not more than its original restriction, plus 1-1/2 inches of water.

3.7.7 Vibration resistance. When tested as specified in 4.5.2.9, the air cleaners shall not leak or show evidence of structural damage or permanent deformation.

3.7.8 Resistance to water washing. When tested as specified in 4.5.2.10, the filter element shall have a restriction recovery within 1 inch of water of rated value, shall have an efficiency of not less than 99 percent, and shall show no evidence of damage or permanent deformation.

3.7.9 Fungus resistance. The air-cleaner element shall not be adversely affected when tested as specified in 4.5.2.11. On completion of the test, the element shall conform to 3.7.2 and 3.7.3.

3.8 Identification marking. The air cleaners shall be identified in accordance with MIL-STD-130, and shall include the maximum rated airflow capacity (see 3.6).

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3.9 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the air cleaner and its components and parts normally painted shall be cleaned, treated, and painted, in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.10 Workmanship. The air cleaner shall be free of burrs, sharp edges, surfaces out of alignment or out of contour (visually apparent), paint sags, runs, blisters, or incomplete coverage, and loose metal particles and other foreign matter.

3.10.1 Sheet metal parts. Sheet metal parts shall be free of pits, laminations, blisters, tears, visual metal thinning at drawn sections, and sharp edges.

3.10.2 Filtering element. The filtering element shall be free of loose material and shall be firmly positioned in its housing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First-produced air cleaner inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First-produced air cleaner inspection.

4.3.1 Examination. The air cleaners shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The air cleaners shall be tested as specified in 4.5.2.1 and 4.5.2.2 and shall be subjected to the tests marked "X" in column 1 of table I in the order listed. Failure of any test shall be cause for rejection.

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TABLE I. Applicable tests.

Test Paragraph	Test	1 First Produced	2 Quality Conformance	3 Control
4.5.2.3	Resistance to air leakage.	X	X	X
4.5.2.4	Airflow restriction.	X	X	X
4.5.2.5	Efficiency at maximum rated airflow.	X	-	X
4.5.2.6	Dust capacity at constant airflow.	X	-	X
4.5.2.7	Service life.	X	-	-
4.5.2.8	Restriction after first cycle of service test.	X	-	-
4.5.2.9	Vibration.	X	-	X
4.5.2.10	Water washing.	X	-	-
4.5.2.11	Fungus.	X	-	-

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105 except as specified in 4.4.1.3.

4.4.1.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S3, except that not less than two shall be selected.

4.4.1.3 Control inspection. Samples for control inspection shall be selected at the rate of one from each 500 air cleaners produced, except that not more than two air cleaners shall be selected in any 30-day period.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 or 4.4.1.3 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be subjected to the tests specified in 4.5.2.1 and 4.5.2.2 and those marked "X" in column 2 of table I. AQL shall be 6.5 percent defective. An additional element shall be used for the dust capacity test at constant airflow (see 4.5.2.6).

4.4.4 Control tests. Samples selected in accordance with 4.4.1.3 shall be subjected to the tests specified in 4.5.2.1 and 4.5.2.2 and as marked "X" in column 3 of table I in the order listed. Failure of any test shall be cause for the Government to refuse to accept air cleaners until it has been proved to the satisfaction of the Government that corrective action has been taken. An additional element shall be used for the dust capacity test (see 4.5.2.6).

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4.5 Inspection procedure.

4.5.1 Examination. The air cleaners shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Service provisions not as specified.
104. Filter element not as specified.
105. Bent or otherwise defective tubular and flanged joints.
106. Major assemblies and attachments not interchangeable.
107. Filter element not secure.
108. Assembly incorrect.
109. Workmanship not as specified.

Minor

201. Identification illegible or missing.
202. Treatment and painting not as specified.
203. Color not as specified.

4.5.2.2 Tests.

4.5.2.1 Test conditions.

4.5.2.1.1 Temperature. The ambient temperature during tests shall be 80° F plus or minus 10° F.

4.5.2.1.2 Relative humidity. The relative humidity during tests shall be 35 percent plus or minus 5 percent.

4.5.2.1.3 Airflow. Specified values of airflow are for the standard atmospheric conditions of 29.92 inches Hg barometric pressure and 80° F ambient temperature. Actual airflow values shall be corrected to standard conditions and shall be within plus or minus 2 percent of specified values.

4.5.2.1.3.1 Pulsation or aspirator airflow. When the dust unloader or aspirator connection type air cleaners are tested, a 10 percent of rated cfm air supply shall be provided for the pulsation or aspirator flow test condition.

4.5.2.1.4 Test dust. The chemical analysis and the particle size distribution by weight of the test dust shall be as specified in SAE J726C.

4.5.2.1.4.1 Dust feeding rate. When the feeding of dust is required as part of any test, the dust shall be fed at an average rate of 0.025 plus or minus 0.005 gram per cubic foot of air, unless otherwise specified herein.

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4.5.2.1.5 Air cleaner preparation. Unless otherwise specified herein, a clean unused filter element shall be installed in the air cleaner for each test (the filter element is not considered "used" after use in a restriction test). Further, the air-cleaner assembly (less filter element) shall be cleaned thoroughly with compressed air if it has been subjected previously to a test in which dust was used.

4.5.2.1.6 Air cleaner mounting position. Unless otherwise specified herein, the air cleaner shall be mounted in a vertical position for all tests specified herein.

4.5.2.2 Test equipment.

4.5.2.2.1 Piezometer, adapter, manometer, and flowmeter. Piezometers conforming to SAE J726C and equal to or the next size larger (maximum of 1/2 inch larger) than the air cleaner inlet shall be installed at the inlet and outlet of the air cleaner. Transition ducts (if required) shall be installed from the flanges of the air cleaner inlet and outlet to the piezometers. Adapters installed after the outlet piezometers shall have sides at angles not more than 7 degrees as measured from the centerline of the transition piece. A smooth approach (ideal flow orifice specified in SAE J726C) shall be installed on the front of the inlet Piezometer tube. A manometer (water type) shall be installed between the two piezometers as shown in figure 2A of SAE J726C. A flowmeter shall be connected to the outlet of the adapter.

4.5.2.2.2 Master filter. The master filter shall consist of one or more layers of 1/2-inch fiberglass batting (Owens Corning Fiberglass Company, FM004, or equal) installed in a housing with the nap side of the fiberglass batting facing upstream. The master filter batting shall be of a size to prevent restriction (pressure drop) to the airflow before the test is completed. A new master filter batting shall be installed if, during any service life or dust capacity test, the air-pressure drop across the master filter increases 10 inches of water above its original restriction. (The face velocity of air against the master filter batting shall be not more than 165 feet per minute. A manometer shall be located in the test setup to measure the pressure drop across the master filter in order to determine when the static pressure across the master filter increases 10 inches of water above its normal restriction). SAE J726 shows the test setup for dust feeding and the dust feeder. The filter housing shall be grounded during tests.

4.5.2.2.3 Preparation of master filter. The fiberglass batting shall be hung in an air oven and dried for 6 hours at a temperature of 220° F to 240° F. This drying time may be reduced and the batting considered as dry if there is no change in weight of the batting at two consecutive 10-minute intervals. A similar drying and weighing procedure shall be conducted after each test.

4.5.2.3 Resistance to air leakage. One of the following air leakage tests shall be used to determine resistance to air leakage at the contractor's option.

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4.5.2.3.1 Air leakage test number 1. Seal the inlet to prevent air leakage at this location. Connect the outlet of the air cleaner to a flowmeter with a maximum flow reading not more than 5 cubic feet of air per minute. Accuracy of flow shall be plus or minus 1/10 of 1 percent of the air cleaner rated flow. Assure tight flowmeter connections. Subject the air cleaner to a vacuum of 50 plus or minus 1-1/2 inches of water as measured at the air cleaner outlet. Evidence of any airflow shall constitute failure of this test.

4.5.2.3.2 Air leakage test number 2. Seal the inlet and the outlet of the clean-air chamber and pressurize the interior of the clean-air chamber to 2 pounds per square inch air pressure through a pipe coupling connected in the drilled and tapped hole in the inlet or outlet sealing plate. Submerge the air cleaner in water. Any evidence of air bubbles escaping from the air cleaner shall constitute failure of this test.

4.5.2.4 Airflow restriction. Conduct this test on a test setup specified in SAE J726. Set the airflow at maximum rated flow and record the restriction. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.5 Efficiency at maximum rated airflow. Prepare the air cleaner as specified in 4.5.2.1.5. Install a master filter as specified in 4.5.2.2.2, start the airflow and adjust to maximum rated flow. If one master filter does not contain sufficient batting area to meet the requirements of 4.5.2.2.2 (165 feet per minute), two or more shall be installed in parallel, as applicable. Feed fine grade test dust as specified in 4.5.2.1.4 into the air cleaner inlet at the rate of 0.025 gram of dust per cubic foot of air until 30 minutes of feeding shall have elapsed. Remove the master filter taking care to retain all entrapped dust. Obtain the efficiency by means of the following formula:

$$\text{Efficiency, percent} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where W_1 = Weight of dust fed into air cleaner inlet.
W_2 = Weight of master filter, outlet conductors (see 6.4.4).
W_3 = Original weight of master filter and outlet conductor.

Note. All dust measurements shall be accurate to the nearest 0.01 gram. Other measurements shall be accurate to nearest 0.01, wherever possible, but never less than to the nearest 0.1 gram. Nonconformance to 3.7.3 shall constitute failure of this test.

4.5.2.6 Dust capacity at constant airflow. After pretest preparation (see 4.5.2.1.5), install a master filter (see 4.5.2.2.2) between the air cleaner and the flowmeter. Weigh the filter element (W_5). Start airflow and adjust to the maximum rated flow of the air cleaner. Measure and record the restriction at start and at 10-minute intervals throughout the test. Feed coarse grade dust, as specified in 4.5.2.1.4, into the air inlet for 30 minutes at the rate of 0.025 gram of dust per cubic foot of air. Weigh the master filter in

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accordance with 4.5.2.2.3. At the end of the first 30 minutes, record the initial specific efficiency. After completing the initial specific efficiency test, replace the fiberglass batting and continue feeding dust to the air cleaner to a 20 inches of water restriction and then weigh and replace the fiberglass batting and the filter element. Determine primary separator efficiency by means of the following formula:

$$\text{Primary separator efficiency, percent} = \frac{W_1 - (W_2-W_3) - (W_4-W_5)}{W_1} \times 100$$

Where W_1, W_2, W_3, and accuracy of dust measurements are as defined in 4.5.2.5.
W_4 = Weight of filter element.
W_5 = Original weight of filter element.

Conduct a second 30-minute specific efficiency test. After completing this second specific efficiency test, replace the fiberglass batting and continue feeding dust to the air cleaner to a 50 inches of water restriction (or 10 hours, whichever occurs first) and weigh and replace the fiberglass batting. Conduct a third and last specific efficiency test for 30 minutes or to a maximum of 70 inches of water, whichever comes first. Nonconformance 3.7.4, 3.7.4.1, or 3.7.4.2 shall constitute failure of this test.

4.5.2.7 Service life. Following completion of the tests specified in 4.5.2.6, empty the air cleaner dust cup, if used, and replace the filter element with a new element. Weigh the new filter element (W_5). Reassemble the cleaner and determine and record the initial restriction. Conduct a constant maximum rated airflow dust capacity test at a dust concentration of 0.100 gram per cubic foot of airflow, to a total restriction of 20 inches of water. Then clean the air cleaner in accordance with 4.5.2.7.1. Repeat the procedure until a total test time of 15 hours for class I, 25 hours for class II, and 30 hours for class III air cleaners are accumulated, or until the frequency of cleanings exceeds that specified in 3.7.5, whichever occurs first. Clean the air cleaner dust cup, if used. Weigh the filter element (W_4) and clean. Determine initial restriction prior to initiation of each dust capacity test. Record the weight of the dust fed for each test. A fiberglass batting shall be installed in the master filter frame at all times. Replace and weigh at the end of each dust capacity cycle, or when the net static pressure rise across the master filter reaches 10 inches of water. Clean the air cleaner element after each cycle in accordance with 4.5.2.7.1. Determine the primary separator efficiency for each test cycle. The arithmetic average of all separator efficiencies shall constitute the total test separator efficiency. Nonconformance to 3.7.5 shall constitute failure of this test.

4.5.2.7.1 Cleaning procedure. Clean the filter element as follows: Use compressed air of approximately 90 psi and blow a stream of dry clean air against the downstream side of the element making certain to avoid damaging

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filter media. Move the nozzle up and down blowing air through the media and slowly rotate element. Continue cleaning for not less than 20 minutes for rated airflows up to 300 cfm and for 30 minutes for rated airflow above 300 CFM. Cleaning may be stopped prior to the times specified herein if no further dust is being removed from the element as determined visibly. After the filter element is clean, reinstall in the air cleaner.

4.5.2.8 Restriction after first cycle of service test. After the first cycle of the service life test has been completed, remove the filtering element and clean in accordance with 4.5.2.7.1. After the filter element is cleaned, reinstall in the air cleaner. Start the airflow and adjust to the maximum rated airflow (cfm). Determine and record the restriction. Nonconformance to 3.7.6 shall constitute failure of this test.

4.5.2.9 Vibration. Mount the air cleaner on a vibrating machine. Subject the air cleaner to a simple harmonic motion having an amplitude of 1/32 inch (1/16 inch total excursion). Uniformly vary the frequency between the limits of 10 and 40 cycles per second. The entire frequency range from 10 to 40 cps and return to 10 cps shall be traversed in 1 minute. Vibrate the air cleaner for 12 hours in a vertical axis followed by 12 hours in a horizontal axis except that 6 hours shall be in one horizontal axis and 6 hours in the other horizontal axis. Then repeat the resistance to air leakage test specified in 4.5.2.3. Nonconformance to 3.7.7 shall constitute failure of this test.

4.5.2.10 Water washing. Air clean (see 4.5.2.7.1) the element used in the dust capacity test (see 4.5.2.6) and immerse it in water for 1 minute. Agitate for 1 minute in the water. Remove and flush with cool water for 1 minute in a direction opposite to element airflow so that the pressure of the water does not damage the element. Allow the water to drain from the element for 5 minutes. Install the element in the air cleaner and operate at 150 percent of maximum rated airflow for 5 minutes. Repeat the water washing and drying procedure eight times. At the end of the last drying period, record the air cleaner restriction at maximum rated airflow. Continue maximum rated flow for 1 hour or until the restriction has recovered to within 1 inch of water of its rated value based on initial restriction from the dust capacity test. Then conduct the efficiency test specified in 4.5.2.5. Examine the filter element. Nonconformance to 3.7.8 shall constitute failure of this test.

4.5.2.11 Fungus. Subject the air cleaner element to method 508, MIL-STD-810. Nonconformance to 3.7.9 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Level S3.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Containers not as specified. Each incorrect container shall be considered one defect.
111.	Box closure for preservation and packing not as specified for level A or level B.
112.	Air cleaners of unlike description packed together for level A or B.
113.	Quantities packed together exceed the weight or size limitation specified for level A or B.
114.	Preservation and packing not in accordance with the referenced document for commercial.
115.	Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each air cleaner shall be placed in a close-fitting box conforming to PPP-B-636, Class-Weather-Resistant, style optional. Box closure and sealing shall be as specified for method V in the appendix to the box specification.

5.1.2 Commercial. Air cleaners shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. Air cleaners of like description, preserved as specified in 5.1, shall be packed as specified for level A, except that boxes may be domestic type or packed. Together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, in quantities not to exceed a gross weight of 250 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Air cleaners of like description, preserved as specified in 5.1, shall be packed as specified for level A, except that boxes may be domestic type or packed together in a close-fitting box conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight or size shall not exceed the limitation of the box, in accordance with the box specification. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Air cleaners, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The air cleaners are intended for use with internal combustion engines for automotive and industrial applications. The application recommendations shown in table II are to be used as a guide in the selection of air cleaners for end items.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Time frame required for submission of first-produced air cleaners and elements, and number required (see 3.2).
d. Maximum rated airflow required when other than as specified (see 3.6).
e. Treatment and painting if other than as specified (see 3.9).
f. Color required (see 3.9).
g. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First-produced air cleaners. Any changes or deviations of production air cleaners from the approved first-produced air cleaners during production will be subject to the approval of the contracting officer. Approval of the first-produced air cleaners will not relieve the contractor of his obligation to furnish air cleaners conforming to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Restriction. The static pressure drop across the air cleaner at the observed airflow, expressed in inches of water.

6.4.2 Accumulative efficiency. The accumulative efficiency is based on the total amount of dust fed and includes one or more specific efficiency tests plus the interval between.

6.4.2.1 Accumulative efficiency basis. The accumulative efficiency to 20 inches water restriction and from 20 to 50 inches water restriction shall be based on the total dust fed in the respective test periods.

6.4.3 Specific efficiency. The efficiency between any specified interval, the interval being defined as either time or restriction rise.

6.4.4 Outlet conductors. Outlet conductors are everything downstream from the air cleaner to the master filter batting.

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6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Review activity:
DLA - CS

User activities:
Army - AT
Navy - MC, AS

Preparing activity:
Army - ME

Project 2940-0117

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TABLE II. Recommended end-item selections.

Class I (8 Hours, 70% Primary Efficiency)	Class II (12 Hours, 70% Primary Efficiency)	Class III (15 Hours, 90% Primary Efficiency)
Batching Plants. 1/	Augers, Earth.	Asphalt plants.
Compressors, Spray-Type, 5 to 25 cfm.	Bins, Aggregate Loading.	Compressors, Reciprocating & Rotary, Rock Drilling & Quarying, 365 to 600 cfm/100 psi.
Curing Machines.	Compressors, Construction Type, 125 to 250 cfm/100 psi.	Cranes & Rough-Terrain & Shovel.
Distributors, Tank-Type.	Cranes, Wheel-Mounted	Crushers & Crushing Plants.
Drier-Mixers, Concrete Mtls.	Crane, Shovel.	Driers, Aggregate.
Finishing Machines, Concrete Paving.	Dehydrators, Sand Compression.	Loaders (Large).
Generator Sets	Ditching Machines.	Rock Drillers, Self-Contained.
Graders, Road. 1/	Dust-Collecting Machines. 2/	Scrapers (Large and Rear Engine).
	Earth-Boring, Rotary.	
Kettle, Heating, Bitumen.	Excavators, Multipurpose.	Scrubbers & Washers, Aggregate.
Mixers. 1/	Fork-Lift Trucks.	Tractors, Wheeled, Industrial.
Pavers. 1/	Loaders.	Tractors, Full-Tracked, Low-Speed.
Pumps, Concrete.	Scrapers, Small.	
Rollers, Motorized.	Tractors, Crawler (less than 100 H.P.).	
Saw. 1/	Tractors, Wheeled, Agricultural.	
Spreaders.	Trucks, Off-Highway.	
Tractors, Farm & Warehouse.		
Trucks, Over-Highway.		
Welders.		
Winches, Drum.		

1/ Class I or Class II, as applicable.

2/ Class II or Class III, as applicable.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-A-52363B Air Cleaners, Intake: Dry-Type (For Internal-Combustion Engine)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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